

Real-Time Intrusion Detection for IoT Networks Using Machine Learning

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Abstract—The usage of Internet of Things (IOT) devices in different fields such as healthcare, smart homes, transportation and industrial automation has made digital networks susceptible to cyber attacks. Although IOT devices are extremely convenient, they usually lack the computing capabilities, do not have standard security regulations and ineffective update options, which is why conventional security practices are not sufficient. The threats have become more complex, and there is the need of a simple, scalable and intelligent security system capable of identifying and preventing attacks fast. The proposed project is an intrusion detection system (IDS) tailored to IOT networks, created and built in real-time, deployed on the machine learning algorithm and network data collected with Wireshark. The IDS is configured on a test network of IOT devices communicating over a local Wi-Fi, which ensures that the system could be tested in real-life.

Index Terms—Internet of Things, Intrusion Detection System, Machine Learning, Wireshark, Network Security, Random Forest, Real-time Monitoring, Cyber Security.

I. INTRODUCTION

The internet of things (IOT) has transformed how we interact with the environment and this is through connecting smart devices which are able to collect, process and share real time data. Since the use of smart homes to automation in industries, IOT is used in many spheres to enhance efficiency, remote monitoring and smart decision making. Nevertheless, with the rapid increase in the number of IOT devices, there is also a greater concern regarding the level of environmental damage, particularly due to the fact that most of these devices use power sources, which are not renewable, and require frequent replacements of the battery. Going green with IOT technologies is one of the significant moves towards the creation of energy-efficient and sustainable digital systems that lessen energy consumption and eliminate electronic waste. These

technologies involve green building and development, renewable energy sources like solar or kinetic energy, low-power technologies and recycling, which would reduce the impact that IOT systems have on the environment. Eco-friendly IOT also contributes towards sustainability as well as bringing new concepts such as energy harvesting and autonomous sensor networks particularly in remote or inaccessible locations. The importance of energy efficiency in IOT systems has been a major design concern over the recent years particularly in devices that use batteries or are located remotely. The conventional IOT gadgets are typically not long lasting since there is limited power, and they need frequent upkeep or replacement of batteries, which are expensive and polluting. The cumulative power consumed is a significant problem since the number of IOT devices that will be utilized is expected to reach billions per the whole world. Energy consumption can be significantly lowered by introducing energy efficient designs and operation like sleep mode, customer designed data transmission schedules and economical hardware parts. More to the point, energy efficiency can enhance the stability of the system because it will not be as dependent on the power grid and will be more independent. The implementation of environmental energy sources such as solar, thermal, vibration or radio frequency enables the device to be run with a little or no reliance on a conventional power source.

II. LITERATURE REVIEW

A. Comparative Analysis of Lightweight ML Techniques Across Diverse Datasets

S. Ismail et al. conducted a comprehensive performance analysis of supervised ML techniques, including Decision Tree (DT), Random Forest (RF), and ensemble methods (Bagging,

Stacking, LightGBM, to identify models lightweight enough for resource-constrained IoT environments. They evaluated metrics crucial for real-time deployment, such as Model Size (MS) and Training Time (TT). They specifically aimed to address the limitations posed by high computational complexity and resource constraints on devices. Furthermore, they performed a cross-dataset transfer learning experiment (TON IoT to WUSTL IIoT-2021) and set up an experiment for real-time network traffic monitoring[1].

B. Feature Selection and Optimization for Resource-Constrained IDS

C. R. Borra et al. developed an intrusion detection method that emphasizes preprocessing with feature selection to improve efficiency. They utilized the Ring-Toss-Game-Based Optimization (RTGBO) algorithm to optimally choose the most relevant features. Their proposed Extreme Learning Machine (ELM) model excelled with the highest accuracy 98.60 the smallest model volume (10 MB), and the fastest prediction time (0.0172 seconds), demonstrating its efficiency and effectiveness compared to other models for fast and lightweight solutions[6].

C. Real-Time DDoS Detection in SDN Environments

K. Hasan et al. specifically focused on developing a real-time DDoS detection system within a Software-Defined Network (SDN) environment using the Mininet simulation tool and Ryu controller. They employed effective feature engineering using Random Forest and PCA, subsequently choosing only 3 key features out of 47. Their results showed that the Decision Tree (DT) model achieved a near-perfect accuracy of 99.97 with optimal model complexity, signifying an effective real-time detection capability[2].

D. Integration of Stream Processing and Middleware for Real-Time Detection:

K. Jaspin et al. addressed challenges posed by the complexity of network traffic by creating a real-time network traffic analysis system. The system combines stream processing platforms like Kafka and Redpanda with a machine learning model within a middleware architecture. They used flow-based values collected from CICFlowmeter as input to a pre-trained Random Forest Classifier. This approach achieved a high accuracy of 99.4 with the Random Forest Classifier, demonstrating that the use of Redpanda alongside ML greatly improves the speed and effectiveness of data processing and real-time classification[14].

E. Edge-Based Model Deployment and Resource Analysis for Anomaly Detection

T. Kindong and S. Iqbal focused on the practical implementation of Supervised ML models at resource-limited edge IoT devices for real-time detection. They tackled the challenge that conventional, computationally costly methods struggle to run on edge hardware. Their study compared traditional ML algorithms like Random Forest, KNN, and Logistic Regression

and explicitly presented findings on the models' memory usage and compute time during both training and real-time prediction, providing crucial metrics for resource-constrained embedded systems.

F. Research Gaps

Despite significant advances, several gaps persist in IoT intrusion detection research:

- **Resource Constraint:** A critical gap exists in developing and validating lightweight ML models optimized for the severe memory and CPU constraints of embedded IoT devices, requiring explicit evaluation of model size and energy use in real-time.
- **Poor Generalization:** Existing ML models lack robustness and frequently fail to generalize effectively across different, heterogeneous IoT domains due to domain shift and variations in traffic characteristics, necessitating a platform-independent solution.
- **Practical Deployment:** Scarcity of research addressing practical deployment challenges including network heterogeneity and scalability.
- **Attack Diversity:** Insufficient coverage of emerging attack vectors specific to IoT ecosystems.

III. PROPOSED METHODOLOGY

IOT Node Identification and System Packet Capture. Installation and identification of the IOT devices in a network is the first step in which they are networked to a central router or access point through a network with the use of Wi-Fi. All packets detected by these IOT devices are gathered based on Wireshark, an effective packet analysis tool. This will guarantee real-time trace of the benign and malicious traffic, as well as communication between the IOT nodes and outside servers. The packet has information like the source IP, destination IP, port, the type of protocol, packet size as well as the time. These packets are later exported in formats such as PCAP or CSV to be analyzed further. The data captured is also identified with the help of such tools as manual actions or Cicflowmeter making it possible to categorize it as the normal or supervised machine learning purposes.

After capturing packets, the second most important task is to extract pertinent features. These are numeric and categorical variables, which are packet length, duration of the flow, protocol, total forward packets, total backward packets, PSH flag numbers, etc. Automatic extraction is done with tools such as Cicflowmeter or with custom python scripts (e.g. Scapy, Pandas). After extraction, the data is processed through various stages such as elimination of the zero values, normalization of the range of features by fair methods such as Min-Max scaling, and label encoding or one-hot encoding of the categorical data. This measurement is necessary to feed the machine learning model with data, which would be stable and produce the correct results.

This is the stage in which the network traffic is classified using machine learning algorithms. Random Forest, Support Vector machine (SVM), decision tree, and XGBoost models are fitted with the dataset, which has been processed. This data will be divided into training and testing data (usually 80-20), and the evaluation of the model will be conducted based on such metrics as the accuracy, precision, the recall, F1-score and ROC-AUC. The cross-validation methods are used e.g., k-folds to make sure that the models are consistent when used in various data splits. The analysis of feature importance is done to identify which indicators of traffic have the most contribution to intrusion detection. The trained model will be incorporated into real-time detection system that will constantly monitor IOT traffic.

When exploited, the system can introduce a code to enable the attacker to carry out further harmful actions, typically targeting data and tasks as the next stage. Infiltration Scene and Logging When exploited, the system may bring in a code which will allow the attacker to perform other malicious activities, often aimed at the data and tasks next. A visualization tool like a Grafana dashboard built on Matplotlib or similar can be added to the system, allowing United Kingdom network administrators to have real-time information. The visualization comprises of time-series graphs of rates of packets, protocol distribution, the number of detection, and threat severity. The cases of discrepancy are indicated in terms of color-code and alert signals. Eventually, all the detection events get stored in a local database or cloud-based database (e.g., SQLite, Firebase) to enable the audit of the detected events and the preservation of the metadata about the detection events to be later analyzed. This allows traceability and makes additional analysis.

Response mechanism provides the network administrators with confirmation or correction of the predictions about the model. The resultant labeled data thus is stored and retrained on with time to attain a more accurate result and minimize the occurrence of a false positive. This adaptive loop allows the system to be flexible to the zero-day attacks as encountered with traffic patterns and new attack vectors. New methods like incremental learning or updating a model can be employed in order to incorporate new knowledge without having to retrain very large models. In edge or embedded systems, the fundamental feature of IOT is energy consumption.

Thus, simplified or lightweight ML models or bare-bone models of more complex ones are executed on platforms such as Raspberry Pi or ESP32. Power consumption is reduced with the assistance of energy-efficient algorithms, as well as low communication frequency. Compression of packets over networks is also used to restrict the size of packet payloads in the process of monitoring, which makes the solution acceptable on IOT networks.

IV. SYSTEM ARCHITECTURE

System architecture is modular layered architecture to facilitate complete threat monitoring, optimized data flow, and scalability of deployment. Architecture is made up of three main layers: IoT Device and Detection Layer, Machine Learning Model Layer, and Network Traffic Analysis Layer connected through cloud-based infrastructure to facilitate optimized communication and data processing.

A. IoT Device and Detection Layer

IoT devices comprise the data source layer that constantly generates network traffic because they are communicating with each other. The layer consists of various devices including sensors, actuators, smart devices, and edge computing nodes. The detection element tracks raw network traffic in real time and also serves as a capture and forwarding module. Packet capture tools like Wireshark or TShark halt data packets, which are forwarded to the next layers for analysis.

B. Machine Learning Model Layer

ML model layer is the analysis engine of the design, it takes labeled and preprocessed network data. Machine learning models trained such as Random Forest, Decision Trees, and Support Vector Machines label traffic activity as normal activity or attempts at an intrusion. The decision-making layer learns continuously through feedback loops and real-time learning pipelines and hence improves with time.

C. Network Traffic Analysis Layer

This layer is accountable for user interface and dashboard integration and communication and also result integration of the ML model. Alert generation, logging, visualization, intrusion reporting, and initiation of automated security response as per system configuration are controlled through functionality. All data traffic, including logarithmic, model training, and alert generation, is controlled through cloud infrastructure or local servers as per deployment needs.

D. Modular Design Benefits

Cloud-ready, modular design enables real-time intrusion detection with scalability across a range of IoT deployments. Decoupling data collection, analysis, and decision making into layers that can be configured together enhances performance and allows easier maintenance and future upgrade. Individual modules can be swapped out independently, such as by an upgrade to more sophisticated ML models, without affecting the system as a whole.

Figure 1 illustrates the complete system architecture showing the interaction between layers and data flow paths.

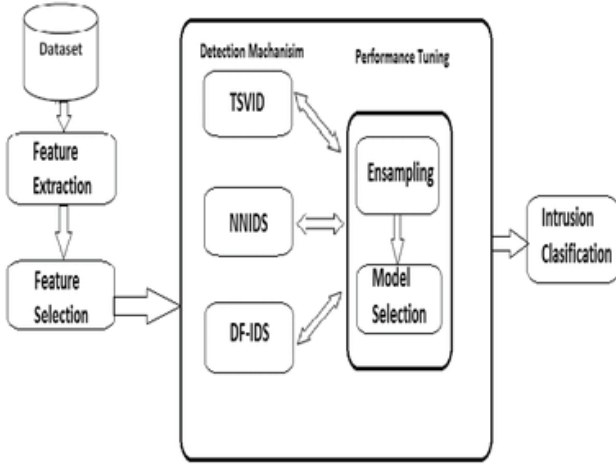


Fig. 1. System Architecture of IoT IDS

Figure 2 presents the system flowchart depicting the sequential processing stages from network traffic capture through alert generation.

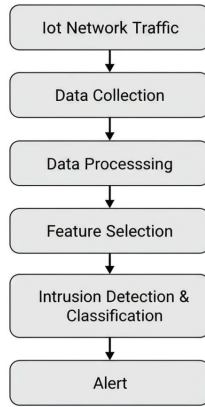


Fig. 2. System Processing Flowchart

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Evaluation Methodology

IOT Node Identification and System Packet Capture. Installation and identification of the IOT devices in a network is the first step in which they are networked to a central router or access point through a network with the use of Wi-Fi. All packets detected by these IOT devices are gathered based on Wireshark, an effective packet analysis tool. This will guarantee real-time trace of the benign and malicious traffic, as well as communication between the IOT nodes and outside servers. The packet has information like the source IP, destination IP, port, the type of protocol, packet size as well as the time. These packets are later exported in formats such as PCAP or csv to be analyzed further. The data captured is also identified with the help of such tools as manual actions or Cflowmeter making it possible to categorize it

as the normal or supervised machine learning purposes. After capturing packets, the second most important task is to extract pertinent features. These are numeric and categorical variables, which are packet length, duration of the flow, protocol, total forward packets, total backward packets, PSH flag numbers, etc.,

TABLE I
EXTRACTED PACKET INFORMATION: TIMESTAMP, PROTOCOL, AND PACKET LENGTH

S.No	Timestamp	Protocol	Length
1	2025-11-13 13:52:58.030443	DNS	75
2	2025-11-13 13:52:58.032185	DNS	91
3	2025-11-13 13:53:03.050190	DNS	85
4	2025-11-13 13:53:03.050313	DNS	85
5	2025-11-13 13:53:03.080993	DNS	113

TABLE II
EXTRACTED NETWORK PACKET INFORMATION: TIMESTAMP AND SOURCE IP

S.No	Timestamp	Source IP
1	2025-11-13 13:52:58.030443	192.168.29.220
2	2025-11-13 13:52:58.032185	192.168.29.1
3	2025-11-13 13:53:03.050190	192.168.29.220
4	2025-11-13 13:53:03.050313	192.168.29.220
5	2025-11-13 13:53:03.080993	192.168.29.1

TABLE III
EXTRACTED NETWORK PACKET INFORMATION: TIMESTAMP AND DESTINATION IP

S.No	Timestamp	Destination IP
1	2025-11-13 13:52:58.030443	192.168.29.1
2	2025-11-13 13:52:58.032185	192.168.29.220
3	2025-11-13 13:53:03.050190	192.168.29.1
4	2025-11-13 13:53:03.050313	192.168.29.1
5	2025-11-13 13:53:03.080993	192.168.29.220

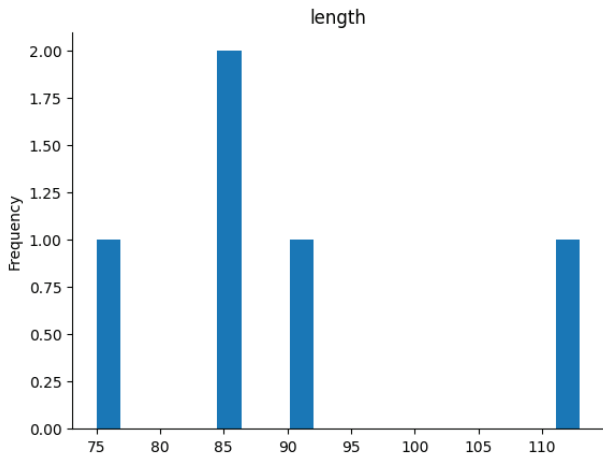


Fig. 3. Histogram of Packet Lengths in Captured Network Traffic

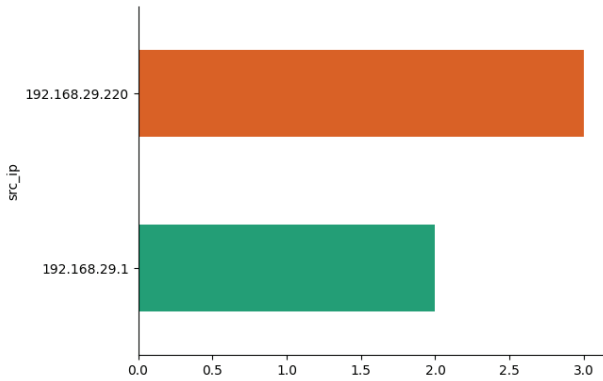


Fig. 4. Frequency Distribution of Source IP Addresses

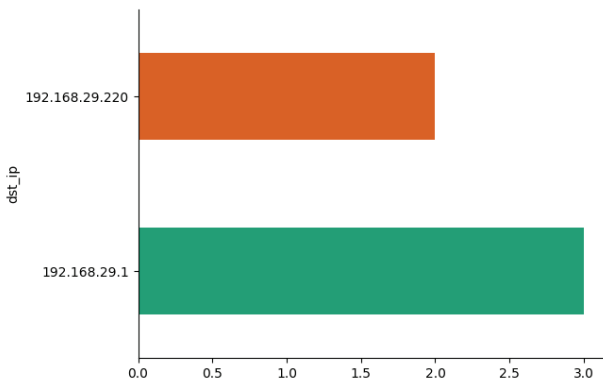


Fig. 5. Frequency Distribution of Destination IP Addresses

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with data, which would be stable and produce the correct results.

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Response mechanism provides the network administrators with confirmation or correction of the predictions about the model. The resultant labeled data thus is stored and retrained on with time to attain a more accurate result and minimize the occurrence of a false positive. This adaptive loop allows the system to be flexible to the zero-day attacks as encountered with traffic patterns and new attack vectors. New methods like incremental learning or updating a model can be employed in order to incorporate new knowledge without having to retrain very large models.

VI. CONCLUSION AND FUTURE WORK

The creation and testing of a real-time infiltration detection system within the IoT network based on machine learning and Wireshark are sufficient evidence to the efficient process of ensuring the safety of the IoT environment against most cyber threats. The system provides a scalable and real time solution by analysing network traffic with Wireshark and applying

machine learning algorithms to detect abnormalities which meets the behavioural anchored rating scale (BARS) model that is fitting in the current IoT network environment.

The process involves several major steps, including the selection of data, data feature, data training and data alerts, and this makes the system effective and precise. According to the results of the experiments and the model analysis, machine learning models (random forest and neural networks) have a better performance and reliability in identifying malicious activities in the IoT traffic. The graphic display through a gradient matrix facilitates the making of a decision based on data. Modularity and real-time processing are also focused on the architecture and are essential in the deployment of live IoT environments.

In general, this project does not only highlight the practical use of AI in infiltration detection, but it also fills the gap between the theoretical study and practise. It plays an important role in the field of IoT-security providing inexpensive, automated, and intelligent intrusion detection system (IDS). The system may be extended or implemented through the addition of deeper learning models, diversifying the data sets or deploying directly on edge computing devices to identify the local threats.

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